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6.5 Angles of Elevation and Depression

The **angle of elevation** is measured upward from a horizontal line of sight to an object above; the **angle of depression** is measured downward from horizontal to an object below. Because the horizontal lines at the observer and at the object are parallel, and the line of sight is a transversal between them, the angle of elevation from the lower point equals the angle of depression from the higher point — they are alternate interior angles. Every elevation/depression problem becomes a right triangle: the horizontal distance is one leg, the height difference is the other leg, and $\tan\theta = \dfrac{\text{height}}{\text{distance}}$ connects them (with sin or cos used when the hypotenuse — the actual line of sight — is involved instead). Some problems require an extra step, such as adding an observer's eye height, or subtracting two elevation readings to isolate the height of just part of an object (like a window). This worksheet models real situations with full multi-step right-triangle trigonometry.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. From 30 ft away, elevation 45°. Find the building height.

2. From 10 m away, elevation 30°. Find the height (exact).

3. From 60 ft up, depression 45°. Find the horizontal distance.

4. Rise 5, run 12. Find the angle of elevation.

5. From 20 ft away, elevation 60°. Find the tree height.

6. Cliff 100 m, depression 30°. Find the distance to the boat.

7. Base 4 ft out, elevation 65°. Find the height on the wall.

8. The angle of elevation from the ground to the top of a tower equals what angle measured from the top?

9. Distance 50, elevation 20°. Find the height.

10. Rise 3 ft, run 15 ft. Find the angle.

11. Height 45, depression 15°. Find the distance.

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12. From 8 m away, elevation $45°$. Find the height.

13. From 40 ft away: elevation to the top of a window is $50°$, to the bottom is $25°$. Find the window's height.

14. A person's eyes are 6 ft above the ground, standing 45 ft from a tower. The angle of elevation from her eyes to the top is $40°$. Find the tower's total height.

15. From the top of a 120 m bridge tower, the angle of depression to a boat is $22°$, and to a buoy farther away is $14°$. Express the distance between the boat and the buoy.

✓ Answer Key

1. 30 ft
2. $\frac{10\sqrt{3}}{3}$ m
3. 60 ft
4. $\tan^{-1}\left(\frac{5}{12}\right)$
5. $20\sqrt{3}$ ft
6. $100\sqrt{3}$ m
7. $4\tan 65°$ ft
8. The angle of depression from the top to the ground
9. $50\tan 20°$
10. $\tan^{-1}\left(\frac{1}{5}\right)$
11. $\frac{45}{\tan 15°}$
12. 8 m
13. $40(\tan 50° - \tan 25°)$ ft
14. $(45\tan 40° + 6)$ ft
15. $\frac{120}{\tan 14°} - \frac{120}{\tan 22°}$ m